Equity Financing Restraint and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Financing Restraint (EFR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on EFR achieves an annualized gross (net) Sharpe ratio of 0.53 (0.47), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 22 (21) bps/month with a t-statistic of 2.93 (2.84), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Growth in book equity, Share issuance (5 year), Net Payout Yield, Change in equity to assets, Asset growth) is 15 bps/month with a t-statistic of 2.24.

1 Introduction

Production-based asset pricing models provide a fundamental framework for understanding cross-sectional variation in expected returns through firms' optimal investment decisions and production technologies. Despite substantial theoretical advances linking firm characteristics to risk premia through production frictions and investment dynamics, empirical evidence connecting financing constraints to expected returns remains mixed. We identify a critical gap in how existing studies measure firms' marginal value of external finance and its implications for cross-sectional pricing. Traditional measures of financial constraints often fail to capture the dynamic trade-offs firms face between internal cash generation and external equity issuance when facing time-varying investment opportunities and adjustment costs.

In a production economy with capital adjustment costs, firms' equity financing decisions reveal important information about their exposure to aggregate productivity shocks and investment frictions. Following the q-theory framework of Cochrane (1991) and Zhang (2005), firms with higher marginal costs of external finance face steeper investment adjustment costs, leading to more volatile marginal products of capital and higher expected returns. The equity financing restraint signal captures firms' revealed preference for internal financing over external equity, which directly maps to the shadow value of internal funds in models with financing frictions (Golan and Miao, 2006). When firms exhibit high equity financing restraint despite growth opportunities, they signal binding financial constraints that amplify their exposure to aggregate productivity shocks through the operating leverage channel documented in Carlson et al. (2004). This mechanism generates time-varying risk premia as constrained firms' investment policies become more procyclical, increasing their systematic risk during economic downturns when the marginal utility of consumption is high. Furthermore, the production-based model of Livdan et al. (2009) demonstrates that financially constrained firms optimally choose more volatile investment paths, generating higher conditional betas with respect to aggregate investment shocks and commanding higher expected returns in equilibrium.

Our empirical analysis reveals that equity financing restraint strongly predicts cross-sectional variation in expected returns. A value-weighted long/short trading strategy based on EFR achieves an annualized gross Sharpe ratio of 0.53, with a monthly average return of 31 basis points and a t-statistic of 4.15. The strategy delivers significant abnormal returns across all standard factor models, with the monthly alpha relative to the Fama-French six-factor model equaling 22 basis points with a t-statistic of 2.93. The economic magnitude of these returns remains substantial after accounting for transaction costs, with net returns of 27 basis points per month and a t-statistic of 3.65. Importantly, the EFR premium persists among the largest stocks in our sample, where firms above the 80th NYSE size percentile generate a monthly return spread of 24 basis points with a t-statistic of 2.68. The strategy's primary risk exposure manifests through a significant loading of 0.34 on the CMA factor with a t-statistic of 6.65, consistent with the interpretation that equity financing restraint captures firms' investment frictions and their resulting exposure to systematic investment shocks. When we control for the six most closely related anomalies from the factor zoo plus the Fama-French six factors, the EFR strategy maintains a significant alpha of 15 basis points per month with a t-statistic of 2.24, demonstrating that our signal contains unique information about expected returns beyond existing measures of equity issuance and asset growth.

This paper makes several contributions to the production-based asset pricing literature by establishing equity financing restraint as a powerful predictor of cross-sectional returns that directly links to firms' production decisions and investment frictions. We extend the findings of Pontiff and Woodgate (2008) and Daniel and Titman (2006) by demonstrating that our measure captures a distinct dimension of financing constraints that existing equity issuance and payout yield variables fail to fully explain. Unlike the mechanical asset growth effect documented in Cooper et al. (2008), our signal specifically identifies firms making active financing choices that reveal their marginal cost of capital and exposure to aggregate productivity shocks. Our results complement Loughran and Ritter (1995) by showing that the negative relation between equity issuance and returns extends beyond SEO announcement effects to a broader measure of financing restraint that captures continuous variation in firms' external finance premium. Methodologically, we contribute to the recent debate on anomaly robustness by demonstrating that the EFR premium survives the comprehensive testing framework of Novy-Marx and Velikov (2023), including controls for transaction costs and related factors that have eliminated many previously documented anomalies. The finding that equity financing restraint maintains explanatory power even among large, liquid stocks suggests that our results reflect fundamental risk compensation rather than market microstructure effects. These findings have important implications for understanding how production frictions and financing constraints jointly determine equilibrium asset prices, providing empirical support for theoretical models that emphasize the role of financial frictions in generating cross-sectional variation in expected returns through firms' endogenous investment and financing policies.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in equity financing activity.construction of our Equity Financing Restraint signal requires two key variables from COMPUSTAT. Common stock

(CSTK) represents the par or stated value of common stock outstanding as reported on the firm's balance sheet. This item reflects the book value of common equity at par, which increases when firms issue new common shares and decreases with share repurchases or retirements. Invested capital (ICAPT) measures the total capital invested in the firm's operations, calculated as the sum of long-term debt, preferred stock, minority interest, and common equity. This variable captures the total permanent capital employed in the business, providing a comprehensive measure of the firm's investment base.construct the Equity Financing Restraint signal by computing the year-over-year change in common stock, scaled by lagged invested capital: (CSTK(t) - CSTK(t-1)) / ICAPT(t-1). This ratio captures the extent of net equity issuance activity relative to the firm's total invested capital base. Positive values indicate net equity issuance, while negative values suggest share repurchases or retirements. By scaling the change in common stock by invested capital, we obtain a measure that is comparable across firms of different sizes and capital structures. We calculate this signal using fiscal year-end values and require non-missing observations for both CSTK and ICAPT in consecutive years. To ensure data quality, we exclude observations where invested capital is zero or negative, as these would result in undefined or economically meaningless ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EFR signal. Panel A plots the time-series of the mean, median, and interquartile range for EFR. On average, the cross-sectional mean (median) EFR is -0.01 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EFR data. The signal's interquartile range spans -0.01 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the EFR signal for the CRSP universe. On average, the EFR signal is available for 6.37% of CRSP names, which on average make up 7.68% of total market capitalization.

4 Does EFR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EFR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EFR portfolio and sells the low EFR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EFR strategy earns an average return of 0.31% per month with a t-statistic of 4.15. The annualized Sharpe ratio of the strategy is 0.53. The alphas range from 0.22% to 0.34% per month and have t-statistics exceeding 2.93 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.34, with a t-statistic of 6.65 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 573 stocks and an average market capitalization of at least $1,481 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 24 bps/month with a t-statistics of 3.14. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for sixteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 20-34bps/month. The lowest return, (20 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.66. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EFR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-three cases.

Table 3 provides direct tests for the role size plays in the EFR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EFR, as well as average returns and alphas for long/short trading EFR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the EFR strategy achieves an average return of 24 bps/month with a t-statistic of 2.68. Among these large cap stocks, the alphas for the EFR strategy relative to the five most common factor models range from 18 to 25 bps/month with t-statistics between 1.93 and 2.73.

5 How does EFR perform relative to the zoo?

Figure 2 puts the performance of EFR in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the EFR strategy falls in the distribution. The EFR strategy's gross (net) Sharpe ratio of 0.53 (0.47) is greater than 92% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EFR strategy (red line).[2] Ignoring trading costs, a \$1 invested in the EFR strategy would have yielded \$7.45 which ranks the EFR strategy in the top 2% across the

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

212 anomalies. Accounting for trading costs, a \$1 invested in the EFR strategy would have yielded \$5.41 which ranks the EFR strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EFR relative to those. Panel A shows that the EFR strategy gross alphas fall between the 65 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EFR strategy has a positive net generalized alpha for five out of the five factor models. In these cases EFR ranks between the 80 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EFR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EFR with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EFR or at least to weaken the power EFR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EFR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EFR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EFR}EFR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EFR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EFR. Stocks are finally grouped into five EFR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EFR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EFR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EFR signal in these Fama-MacBeth regressions exceed 2.83, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on EFR is 1.47.

stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

Similarly, Table 5 reports results from spanning tests that regress returns to the EFR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EFR strategy earns alphas that range from 15-25bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.99, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EFR trading strategy achieves an alpha of 15bps/month with a t-statistic of 2.24.

7 Does EFR add relative to the whole zoo?

Finally, we can ask how much adding EFR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EFR signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes EFR grows to $2676.66.

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EFR is available.

8 Conclusion

This study provides compelling evidence that Equity Financing Restraint (EFR) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EFR-based trading strategies generate substantial risk-adjusted returns even after accounting for transaction costs and controlling for established risk factors.

The key findings reveal that a value-weighted long/short strategy based on EFR delivers impressive performance metrics, with an annualized net Sharpe ratio of 0.47 and statistically significant alpha of 21 basis points per month relative to the Fama-French five-factor model augmented with momentum (t-statistic = 2.84). Notably, the signal maintains its predictive power even when controlling for six closely related equity financing and growth-based factors, generating a monthly alpha of 15 basis points (t-statistic = 2.24). This resilience suggests that EFR captures unique information about firm fundamentals and market mispricing not fully explained by existing factors in the literature.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, EFR contributes to our understanding of how corporate financing decisions influence asset prices and provides further evidence of market inefficiencies related to equity issuance activities. For practitioners, the signal's robust performance after transaction costs suggests potential implementation value in quantitative equity strategies, particularly given its orthogonality to related financing and growth factors.

However, several limitations warrant consideration. First, while our analysis accounts for transaction costs, real-world implementation challenges such as market impact and capacity constraints may further erode returns, particularly for smaller, less liquid stocks. Second, the sample period and market conditions examined may not fully capture the signal's performance across different market regimes or economic cycles. Third, the economic mechanism driving EFR's predictive power requires further theoretical development to fully understand why markets appear to systematically misprice firms based on their equity financing restraint.

Future research should explore several promising avenues. First, investigating the time-varying nature of EFR's predictive power could reveal whether the signal's efficacy depends on market conditions, investor sentiment, or macroeconomic factors. Second, examining EFR's performance in international markets would test the generalizability of our findings beyond U.S. equities. Third, decomposing EFR into its constituent components might identify which specific aspects of equity financing restraint drive the return predictability. Finally, exploring potential interactions between EFR and other corporate finance variables could enhance our understanding of how financing decisions collectively influence asset prices. Such investigations would not only strengthen the empirical foundation of EFR as a pricing factor but also contribute to the broader literature on the intersection of corporate finance and asset pricing.

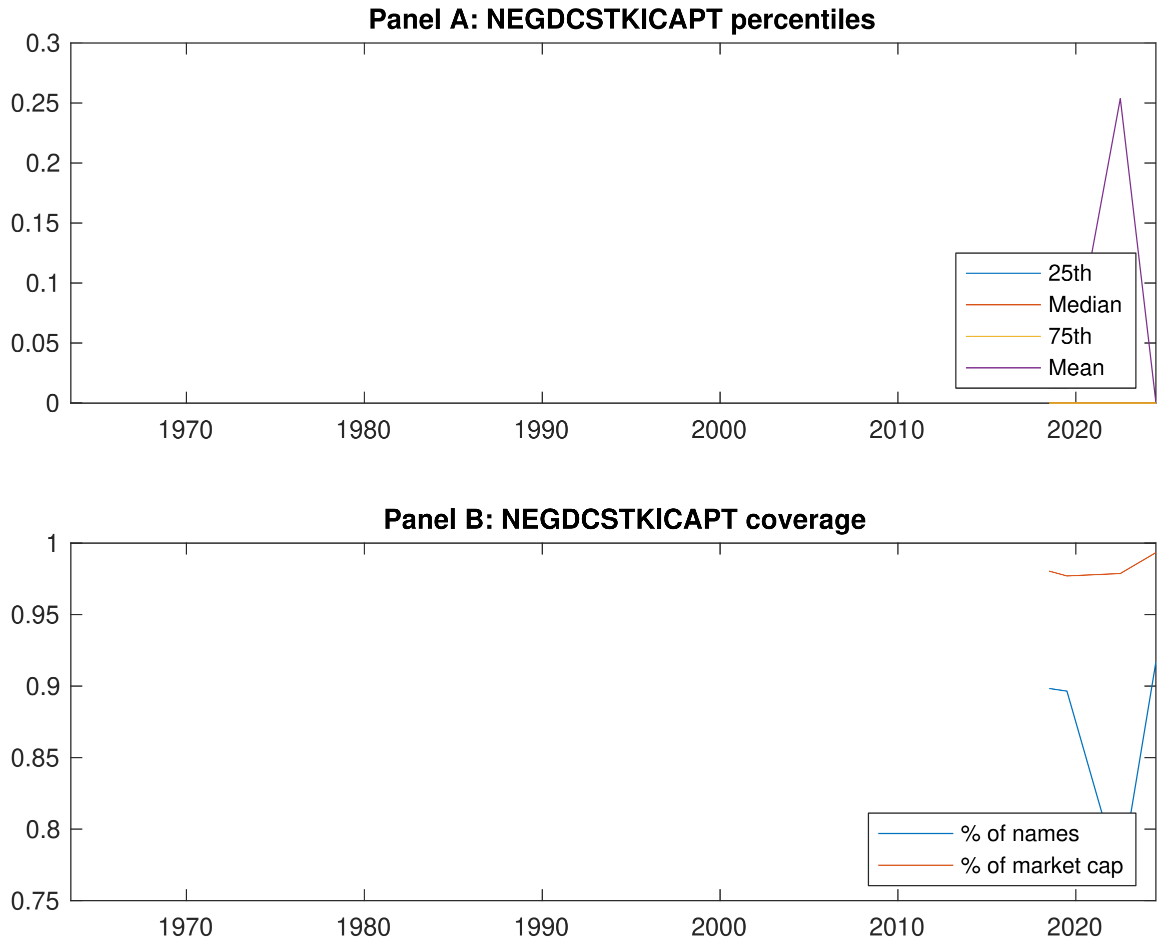


Figure 1: Times series of EFR percentiles and coverage.
This figure plots descriptive statistics for EFR. Panel A shows cross-sectional percentiles of EFR over the sample. Panel B plots the monthly coverage of EFR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EFR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EFR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.43	0.55	0.62	0.67	0.74	0.31
	[2.56]	[3.05]	[3.47]	[4.10]	[4.68]	[4.15]
α_{CAPM}	-0.14	-0.07	0.01	0.11	0.20	0.34
	[-2.79]	[-1.62]	[0.13]	[2.34]	[4.35]	[4.56]
α_{FF3}	-0.14	-0.06	0.02	0.07	0.16	0.30
	[-2.81]	[-1.37]	[0.51]	[1.61]	[3.57]	[4.02]
α_{FF4}	-0.12	-0.03	0.04	0.03	0.15	0.27
	[-2.35]	[-0.65]	[0.82]	[0.73]	[3.38]	[3.59]
α_{FF5}	-0.16	0.00	0.04	0.01	0.07	0.24
	[-3.24]	[0.01]	[0.78]	[0.12]	[1.64]	[3.16]
α_{FF6}	-0.15	0.02	0.05	-0.02	0.07	0.22
	[-2.85]	[0.50]	[1.02]	[-0.49]	[1.71]	[2.93]
Panel B: Fama and French (2018) 6-factor model loadings for EFR-sorted portfolios						
β_{MKT}	0.97	1.02	1.02	1.01	0.97	0.00
	[78.87]	[94.38]	[88.97]	[91.17]	[93.02]	[0.01]
β_{SMB}	-0.02	0.02	0.04	-0.07	-0.01	0.01
	[-0.86]	[1.42]	[2.53]	[-4.55]	[-0.42]	[0.34]
β_{HML}	0.05	-0.01	-0.07	0.08	0.02	-0.03
	[2.13]	[-0.39]	[-3.20]	[3.91]	[1.03]	[-0.85]
β_{RMW}	0.13	-0.11	-0.03	0.10	0.13	-0.00
	[5.34]	[-5.31]	[-1.56]	[4.46]	[6.13]	[-0.10]
β_{CMA}	-0.11	-0.10	0.01	0.15	0.24	0.34
	[-3.04]	[-3.23]	[0.28]	[4.80]	[7.92]	[6.65]
β_{UMD}	-0.03	-0.03	-0.02	0.04	-0.01	0.02
	[-2.14]	[-2.93]	[-1.52]	[3.67]	[-0.59]	[1.12]
Panel C: Average number of firms (n) and market capitalization (me)						
n	759	685	573	674	736	
me ($\$10^6$)	1751	1481	2106	2276	2519	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the EFR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.31	0.34	0.30	0.27	0.24	0.22
			[4.15]	[4.56]	[4.02]	[3.59]	[3.16]	[2.93]
Quintile	NYSE	EW	0.53	0.60	0.53	0.45	0.41	0.36
			[8.77]	[10.30]	[9.67]	[8.39]	[7.82]	[6.87]
Quintile	Name	VW	0.31	0.33	0.30	0.27	0.24	0.23
			[4.09]	[4.29]	[3.84]	[3.42]	[3.17]	[2.93]
Quintile	Cap	VW	0.24	0.26	0.23	0.19	0.21	0.18
			[3.14]	[3.43]	[3.07]	[2.47]	[2.75]	[2.31]
Decile	NYSE	VW	0.26	0.28	0.21	0.17	0.21	0.18
			[2.94]	[3.07]	[2.41]	[1.91]	[2.39]	[2.00]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.27	0.30	0.26	0.25	0.22	0.21
			[3.65]	[4.06]	[3.54]	[3.32]	[2.96]	[2.84]
Quintile	NYSE	EW	0.34	0.42	0.35	0.31	0.23	0.21
			[5.00]	[6.44]	[5.70]	[5.12]	[4.05]	[3.67]
Quintile	Name	VW	0.27	0.29	0.25	0.24	0.22	0.21
			[3.59]	[3.81]	[3.36]	[3.13]	[2.92]	[2.79]
Quintile	Cap	VW	0.20	0.22	0.19	0.17	0.18	0.17
			[2.66]	[2.96]	[2.59]	[2.26]	[2.47]	[2.24]
Decile	NYSE	VW	0.22	0.23	0.17	0.15	0.18	0.16
			[2.46]	[2.59]	[1.98]	[1.71]	[2.01]	[1.83]

Table 3: Conditional sort on size and EFR

This table presents results for conditional double sorts on size and EFR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EFR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EFR and short stocks with low EFR .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		EFR Quintiles					EFR Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.48	0.65	0.87	0.89	1.02	0.56	0.61	0.56	0.51	0.45	0.42
		[1.92]	[2.57]	[3.46]	[3.62]	[4.46]	[7.07]	[7.75]	[7.33]	[6.56]	[5.92]	[5.38]
	(2)	0.56	0.71	0.90	0.89	0.96	0.41	0.46	0.38	0.33	0.32	0.29
		[2.46]	[3.01]	[3.82]	[3.99]	[4.49]	[4.81]	[5.40]	[4.62]	[3.95]	[3.90]	[3.43]
	(3)	0.56	0.64	0.76	0.84	0.91	0.35	0.38	0.33	0.29	0.29	0.26
		[2.78]	[3.00]	[3.46]	[4.14]	[4.70]	[4.76]	[5.04]	[4.45]	[3.87]	[3.86]	[3.44]
	(4)	0.51	0.61	0.78	0.77	0.81	0.30	0.34	0.27	0.26	0.13	0.13
		[2.68]	[3.10]	[3.84]	[4.12]	[4.50]	[4.03]	[4.53]	[3.83]	[3.59]	[1.84]	[1.86]
	(5)	0.45	0.48	0.50	0.56	0.69	0.24	0.25	0.22	0.18	0.21	0.18
		[2.74]	[2.66]	[2.89]	[3.41]	[4.36]	[2.68]	[2.73]	[2.41]	[1.93]	[2.28]	[1.93]

Panel B: Portfolio average number of firms and market capitalization

		EFR Quintiles Average n					EFR Quintiles Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	382	383	382	381	374	31	33	41	30	29
	(2)	109	107	108	108	106	57	56	58	56	56
	(3)	78	78	77	77	78	98	95	98	100	100
	(4)	65	65	65	65	66	204	205	216	212	218
	(5)	60	60	60	60	60	1478	1429	1664	1677	1898

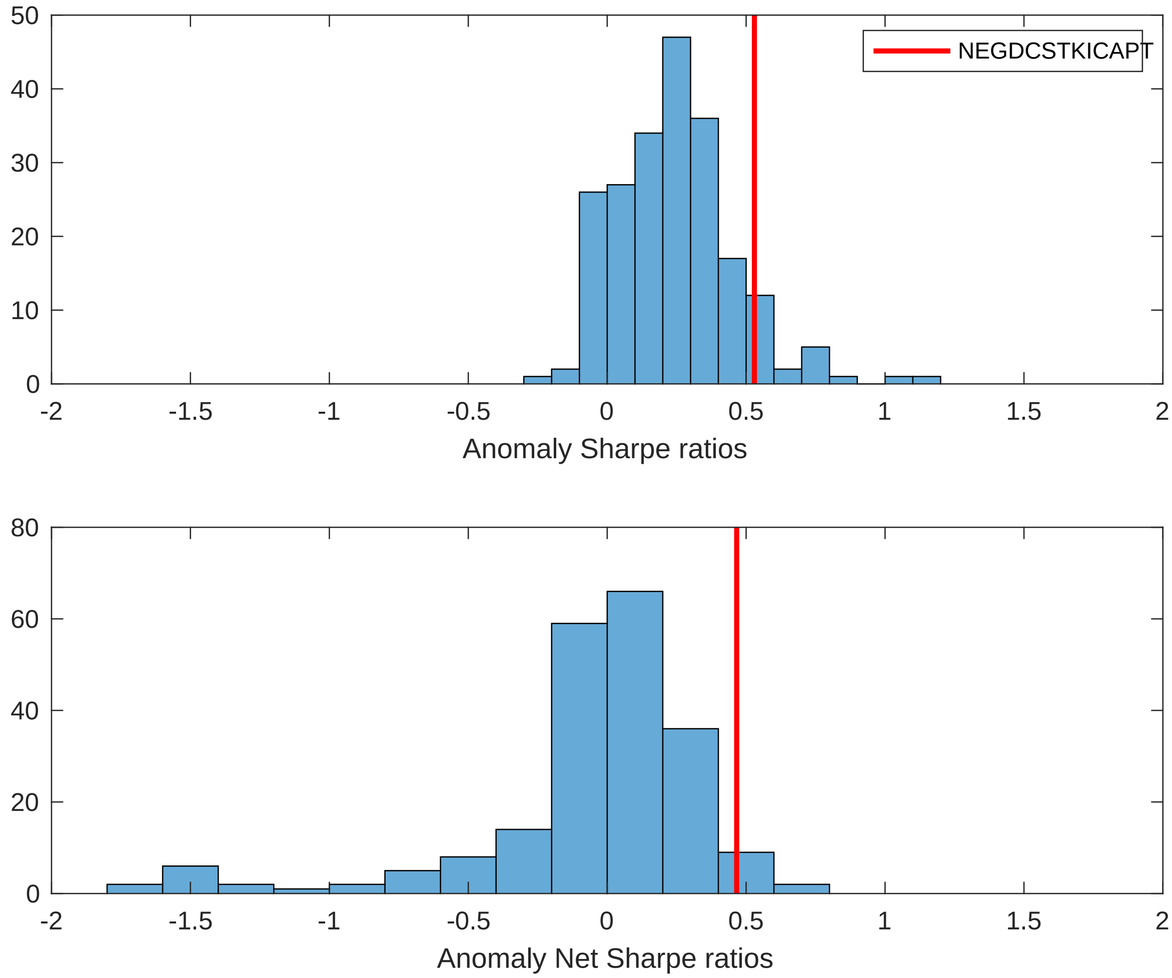


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EFR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

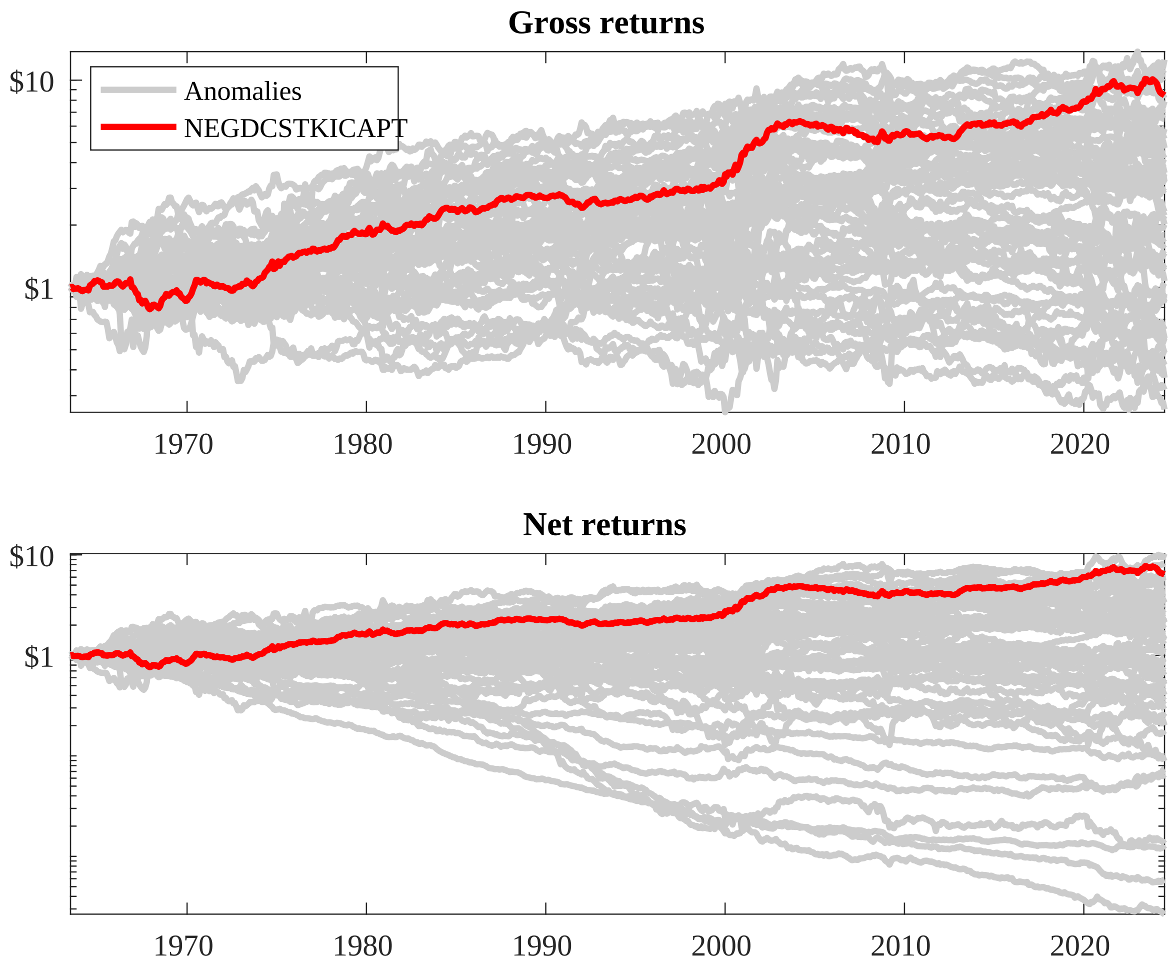
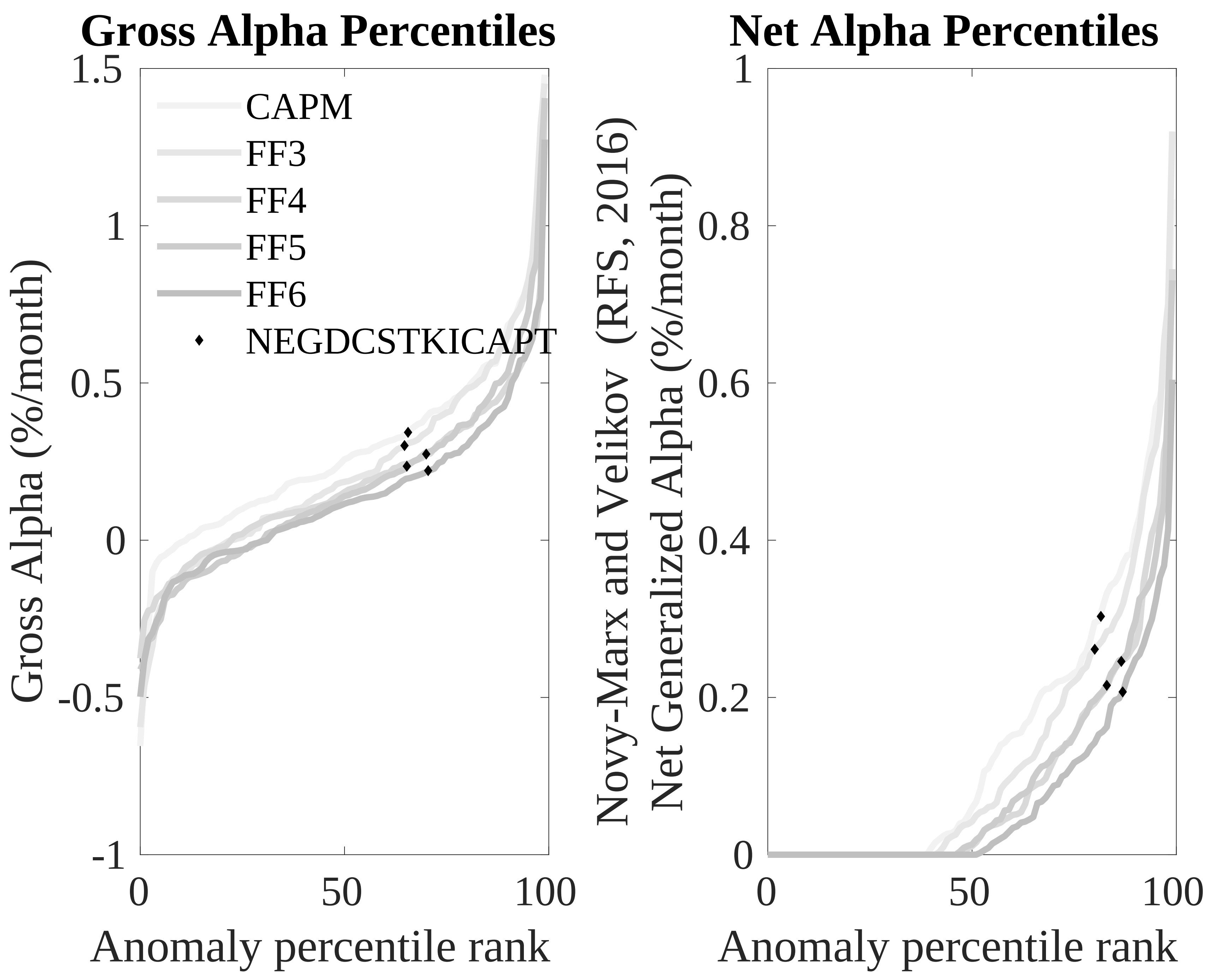
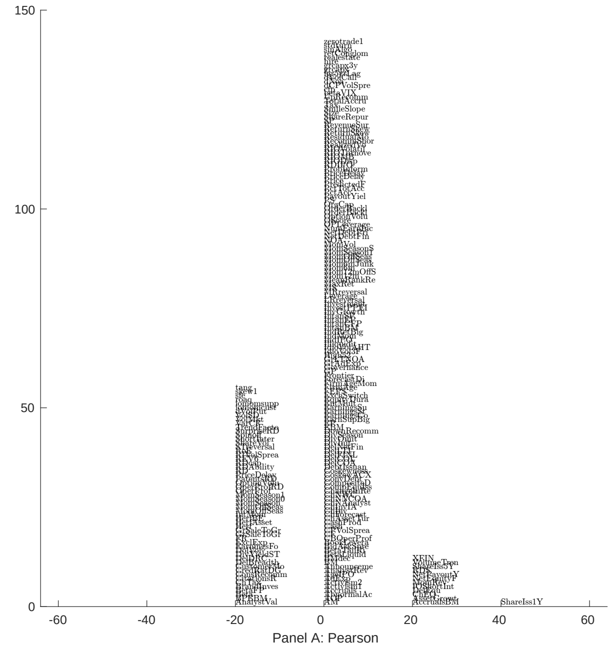


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EFR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDCSTKICAPT
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100



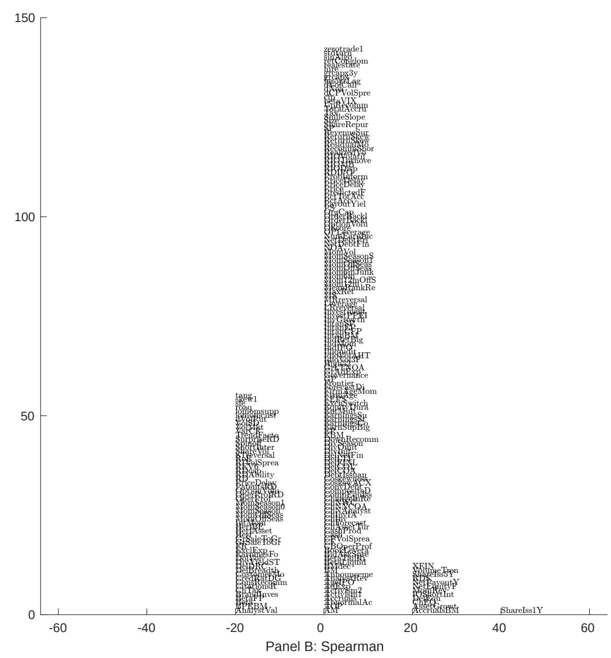


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with EFR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

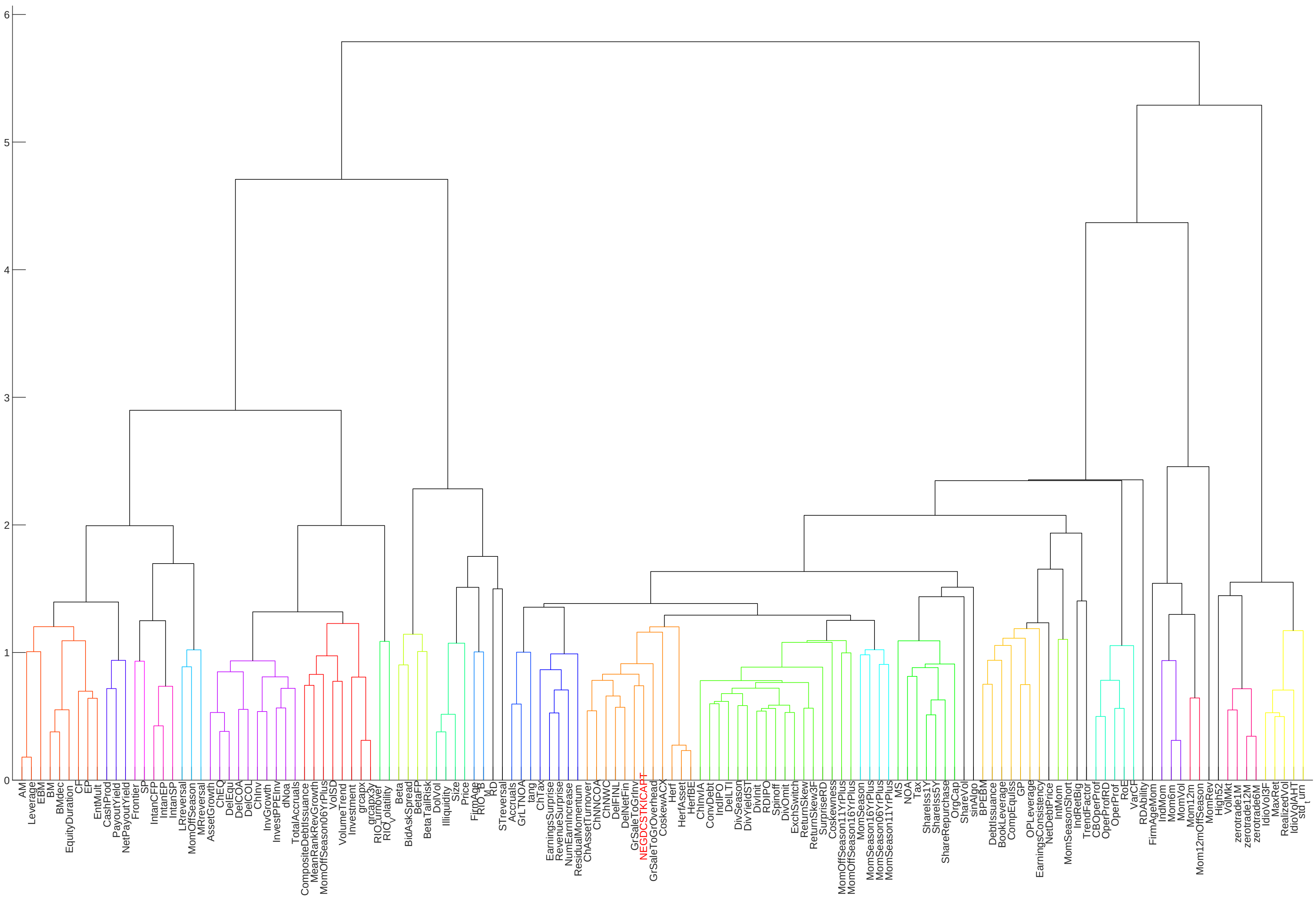


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

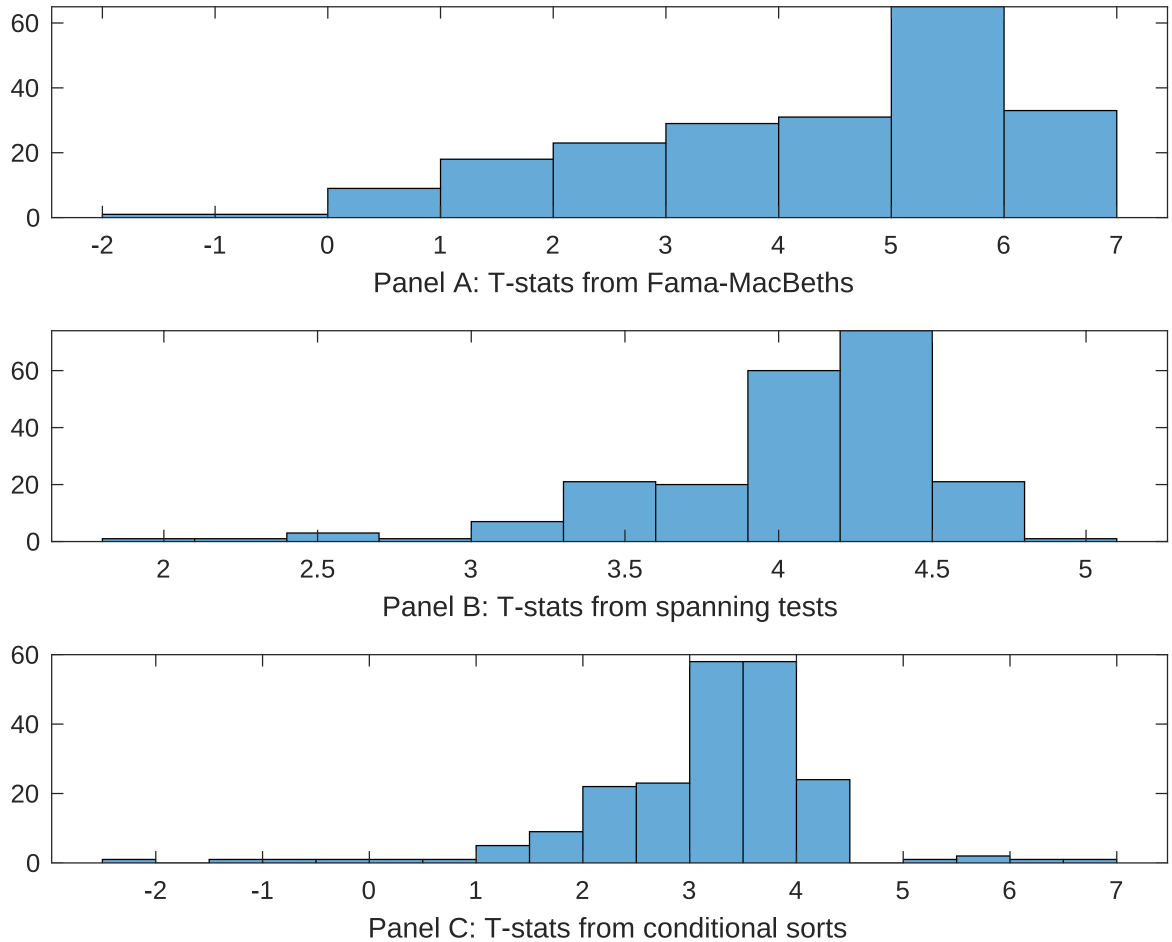


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of EFR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EFR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EFR} EFR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EFR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EFR. Stocks are finally grouped into five EFR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EFR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on EFR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EFR} EFR_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Share issuance (1 year), Growth in book equity, Share issuance (5 year), Net Payout Yield, Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13	0.16	0.13	0.12	0.12	0.13	0.12
	[6.07]	[6.82]	[6.45]	[5.43]	[5.71]	[6.17]	[3.91]
EFR	0.40	0.36	0.36	0.30	0.44	0.37	0.15
	[4.82]	[3.88]	[4.39]	[2.83]	[4.82]	[3.99]	[1.47]
Anomaly 1	0.13						0.66
	[5.87]						[2.40]
Anomaly 2		0.37					-0.13
		[3.17]					[-0.59]
Anomaly 3			0.26				0.24
			[4.77]				[4.27]
Anomaly 4				0.21			0.93
				[1.85]			[0.88]
Anomaly 5					0.11		-0.64
					[3.08]		[-0.11]
Anomaly 6						0.92	0.65
						[7.86]	[5.45]
# months	715	720	715	715	720	720	715
$\bar{R}^2$(%)	0	0	0	1	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EFR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EFR} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Share issuance (1 year), Growth in book equity, Share issuance (5 year), Net Payout Yield, Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.20	0.20	0.15	0.25	0.22	0.21	0.15
	[2.76]	[2.83]	[1.99]	[3.36]	[2.94]	[2.85]	[2.24]
Anomaly 1	30.08						15.75
	[8.25]						[3.61]
Anomaly 2		31.95					37.03
		[8.21]					[6.67]
Anomaly 3			28.73				12.97
			[7.40]				[3.00]
Anomaly 4				17.51			5.57
				[6.16]			[1.83]
Anomaly 5					13.68		-9.63
					[3.70]		[-1.89]
Anomaly 6						-3.55	-21.73
						[-0.80]	[-4.55]
mkt	0.93	1.25	1.48	2.91	-0.05	0.20	3.33
	[0.54]	[0.73]	[0.85]	[1.62]	[-0.03]	[0.11]	[1.96]
smb	3.19	0.93	3.91	4.84	1.83	2.29	5.15
	[1.29]	[0.38]	[1.55]	[1.88]	[0.71]	[0.88]	[2.10]
hml	-2.68	-5.72	0.44	-8.67	-3.59	-1.61	-6.15
	[-0.81]	[-1.71]	[0.13]	[-2.42]	[-1.04]	[-0.46]	[-1.82]
rmw	-16.68	1.18	-10.68	-10.58	0.92	-0.28	-15.74
	[-4.29]	[0.35]	[-2.92]	[-2.79]	[0.26]	[-0.08]	[-3.90]
cma	14.90	2.53	19.63	18.81	19.71	37.77	13.12
	[2.79]	[0.41]	[3.75]	[3.43]	[3.15]	[5.09]	[1.80]
umd	0.81	1.62	0.30	3.58	2.52	2.11	-0.51
	[0.47]	[0.95]	[0.18]	[2.05]	[1.43]	[1.18]	[-0.30]
# months	728	732	728	728	732	732	728
$\bar{R}^2(\%)$	18	17	16	14	11	10	26

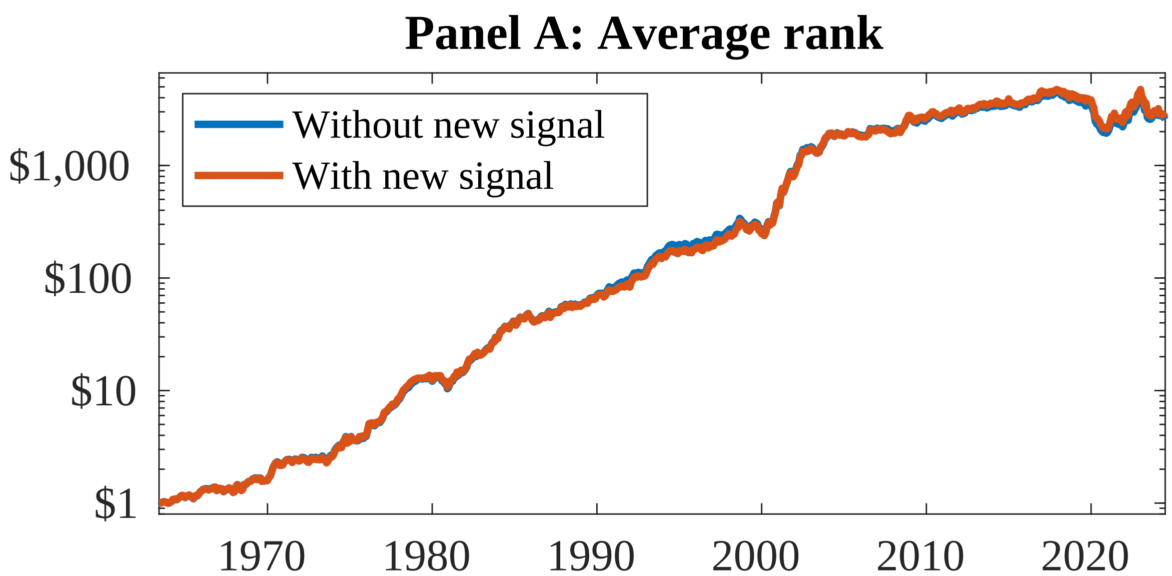


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EFR. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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